1 Devices in the Topology

From the diagram there are:

Device	Quantity
Router	1
Switch 3560 (Central)	1
Switch 2950 (Distribution)	2
Switch 2960 (Access)	2
PCs	8

Rename switches for clarity:

Position	Name
Top switch under router	**SW-Core (Central Switch)**
Left middle	**SW-1**
Right middle	**SW-2**
Bottom left	**SW-Access-L**
Bottom right	**SW-Access-R**

---

2 Physical Connections

Router to Central Switch

```
R1 Gig0/1 → SW-Core Fa0/1
```

Central Switch to Distribution Switches

```
SW-Core Gi0/1 → SW-1 Gi0/1
SW-Core Gi0/2 → SW-2 Gi0/1
```

Distribution to Access Switches

```
SW-1 Fa0/23 → SW-Access-L Fa0/23
SW-1 Fa0/24 → SW-Access-R Fa0/24

SW-2 Fa0/23 → SW-Access-R Fa0/23
SW-2 Fa0/24 → SW-Access-L Fa0/24
```

This creates **redundant paths for STP**.

PCs to Access Switches

Left switch

```
C1 → Fa0/1
C2 → Fa0/2
C3 → Fa0/3
C4 → Fa0/4
```

Right switch

```
D1 → Fa0/1
D2 → Fa0/2
D3 → Fa0/3
D4 → Fa0/4
```

---

3 Configure Router IP

Open **R1 CLI**

```
enable
configure terminal
```

Configure LAN interface

```
interface g0/1
ip address 192.168.1.1 255.255.255.0
no shutdown
```

Configure serial

```
interface s0/0/0
ip address 209.165.200.1 255.255.255.0
no shutdown
```

---

4 Assign Central Switch as Root Bridge

You must configure this on **SW-Core**.

Open **SW-Core CLI**

```
enable
configure terminal
```

Set root bridge

```
spanning-tree vlan 1 root primary
```

This forces this switch to become **STP root bridge**.

Verify

```
show spanning-tree
```

---

5 Enable STP Security (Prevent Manipulation)

We use:

- **BPDU Guard**
- **Root Guard**
- **PortFast**

On Access Switches (SW-Access-L and SW-Access-R)

Open CLI

```
enable
configure terminal
```

Enable PortFast globally

```
spanning-tree portfast default
```

Enable BPDU Guard

```
spanning-tree portfast bpduguard default
```

This prevents **fake switches sending BPDUs**.

On Distribution Switches (SW-1 and SW-2)

Enable **Root Guard** on ports toward access switches.

Example on SW-1:

```
enable
configure terminal
interface fa0/23
spanning-tree guard root
interface fa0/24
spanning-tree guard root
```

Do same on **SW-2**

```
interface fa0/23
spanning-tree guard root
interface fa0/24
spanning-tree guard root
```

---

6 Enable Port Security (Prevent CAM Table Overflow)

Port security must be configured **on PC ports**.

Only on **Access switches**.

On SW-Access-L

```
enable
configure terminal
```

Configure ports

```
interface range fa0/1 - 4
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
```

On SW-Access-R

Same configuration

```
interface range fa0/1 - 4
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
```

---

7 Configure PC IP Addresses

Use the addressing table.

C1

```
IP 10.1.1.10
Mask 255.255.255.0
Gateway 10.1.1.1
```

C2

```
10.1.1.11
255.255.255.0
10.1.1.1
```

C3

```
10.1.1.12
255.255.255.0
10.1.1.1
```

C4

```
10.1.1.13
255.255.255.0
10.1.1.1
```

D1

```
10.1.1.14
255.255.255.0
10.1.1.1
```

D2

```
10.1.1.15
255.255.255.0
10.1.1.1
```

D3

```
10.1.1.16
255.255.255.0
10.1.1.1
```

D4

```
10.1.1.17
255.255.255.0
10.1.1.1
```

---

8 Verification Commands

Check STP root

```
show spanning-tree
```

Check port security

```
show port-security
```

Check learned MAC addresses

```
show port-security address
```

---

9 Testing

From **PC C1**

```
ping 10.1.1.17
```

Should reach **D4**.

Likely Viva Questions

What is STP?
STP (Spanning Tree Protocol) prevents **network loops in Layer 2 networks**.

What is Root Bridge?
Root bridge is the **central reference switch used by STP to calculate paths**.

What is BPDU Guard?
BPDU Guard disables a port if it receives **unexpected BPDU packets**.

What is CAM table attack?
It floods switches with **fake MAC addresses**, causing switch to broadcast traffic.

What is Port Security?
Port security limits **number of MAC addresses allowed on a port**.